Lab 2 – Containerization & Kubernetes Orchestration

Task 2: Manage Docker Containers and Images

Name: Prince Raj

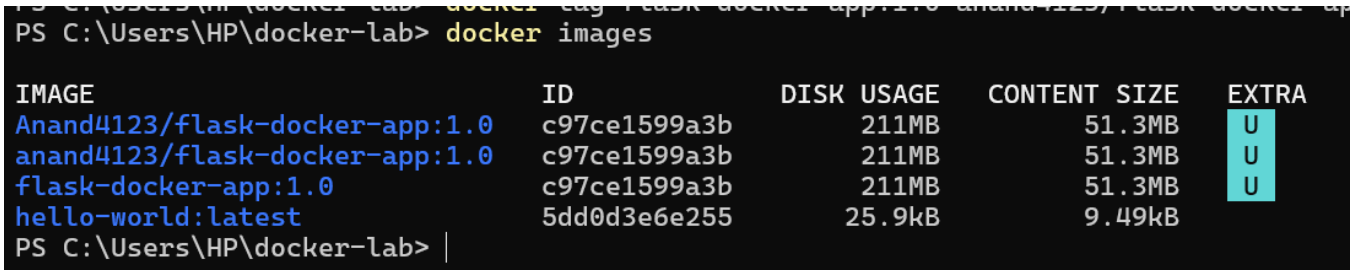
1. Docker Container Listing

```
PS C:\Users\HP\docker-lab> docker ps
CONTAINER ID   IMAGE                 COMMAND           CREATED         STATUS          PORTS                                         NAMES
8addceeb23fe   flask-docker-app:1.0   "python app.py"   7 minutes ago   Up 14 seconds   0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp   flask-container
PS C:\Users\HP\docker-lab> docker ps
CONTAINER ID   IMAGE                 COMMAND           CREATED          STATUS          PORTS                                         NAMES
8addceeb23fe   flask-docker-app:1.0   "python app.py"   25 minutes ago   Up 18 minutes   0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp   flask-container
PS C:\Users\HP\docker-lab> docker ps -a
CONTAINER ID   IMAGE                 COMMAND           CREATED          STATUS                      PORTS                                         NAMES
8addceeb23fe   flask-docker-app:1.0   "python app.py"   26 minutes ago   Up 19 minutes               0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp   flask-con
tainer
995bf9fecf48   hello-world           "/hello"          59 minutes ago   Exited (0) 59 minutes ago                                                 vigilant_
wozniak
```

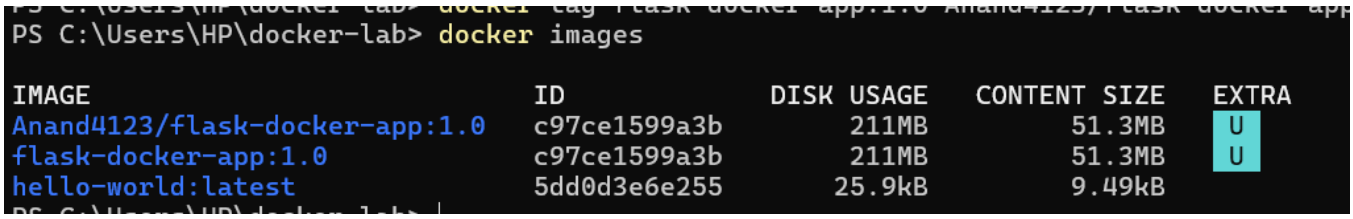
2. Container Stop, Start and Remove

```
PS C:\Users\HP\docker-lab> docker stop flask-container
flask-container
PS C:\Users\HP\docker-lab> docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
PS C:\Users\HP\docker-lab> docker ps -a
CONTAINER ID   IMAGE                  COMMAND           CREATED             STATUS                              PORTS     NAMES
8addceeb23fe   flask-docker-app:1.0   "python app.py"   28 minutes ago      Exited (137) About a minute ago             flask-container
995bf9fecf48   hello-world            "/hello"          About an hour ago   Exited (0) About an hour ago                vigilant_wozniak
PS C:\Users\HP\docker-lab> docker start flask-container
flask-container
PS C:\Users\HP\docker-lab> docker ps
CONTAINER ID   IMAGE                  COMMAND           CREATED          STATUS          PORTS                                       NAMES
8addceeb23fe   flask-docker-app:1.0   "python app.py"   30 minutes ago   Up 23 seconds   0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp   flask-container
PS C:\Users\HP\docker-lab> docker ps -a
CONTAINER ID   IMAGE                  COMMAND           CREATED             STATUS                         PORTS                                       NAM
ES
8addceeb23fe   flask-docker-app:1.0   "python app.py"   31 minutes ago      Up About a minute              0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp   fla
sk-container
995bf9fecf48   hello-world            "/hello"          About an hour ago   Exited (0) About an hour ago                                               vig
ilant_wozniak
PS C:\Users\HP\docker-lab> docker rm vigilant_wozniak
vigilant_wozniak
PS C:\Users\HP\docker-lab> docker ps -a
CONTAINER ID   IMAGE                  COMMAND           CREATED          STATUS         PORTS                                       NAMES
8addceeb23fe   flask-docker-app:1.0   "python app.py"   32 minutes ago   Up 2 minutes   0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp   flask-container
PS C:\Users\HP\docker-lab> docker rm vigilant_wozniak
Error response from daemon: No such container: vigilant_wozniak
PS C:\Users\HP\docker-lab> docker ps -a
CONTAINER ID   IMAGE                  COMMAND           CREATED          STATUS         PORTS                                       NAMES
8addceeb23fe   flask-docker-app:1.0   "python app.py"   32 minutes ago   Up 2 minutes   0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp   flask-container
```

3. Docker Image Listing



4. Docker Image Tagging



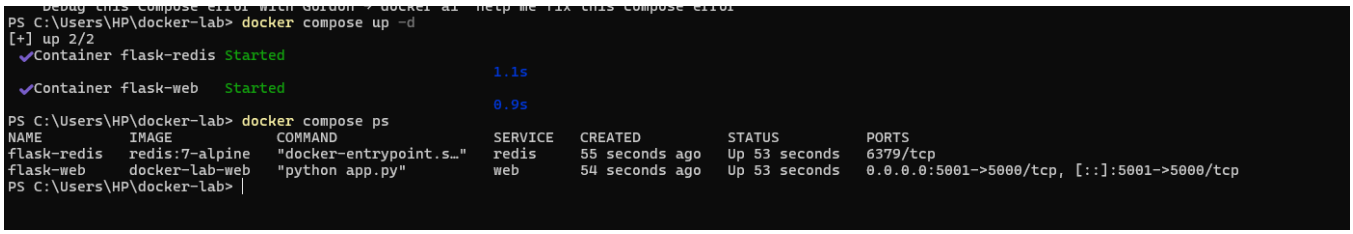
5. Docker Hub Image Push

```
PS C:\Users\HP\docker-lab> docker push anand4123/flask-docker-app:1.0
The push refers to repository [docker.io/anand4123/flask-docker-app]
87e1b7cce023: Pushed
6310eb16bf42: Pushed
7ea3ea474f31: Pushed
44136fa355b3: Pushed
c86306e32cd0: Pushed
a14578096eda: Pushed
7eef5e4f9f95: Pushed
59cfb10edb51: Pushed
23ecc929b5d5: Pushed
0318cf563730: Pushed
1.0: digest: sha256:c97ce1599a3bbefa00fb3cbec2779f39941f6d002609359e1d8e0f0509ada2a0 size: 856
```

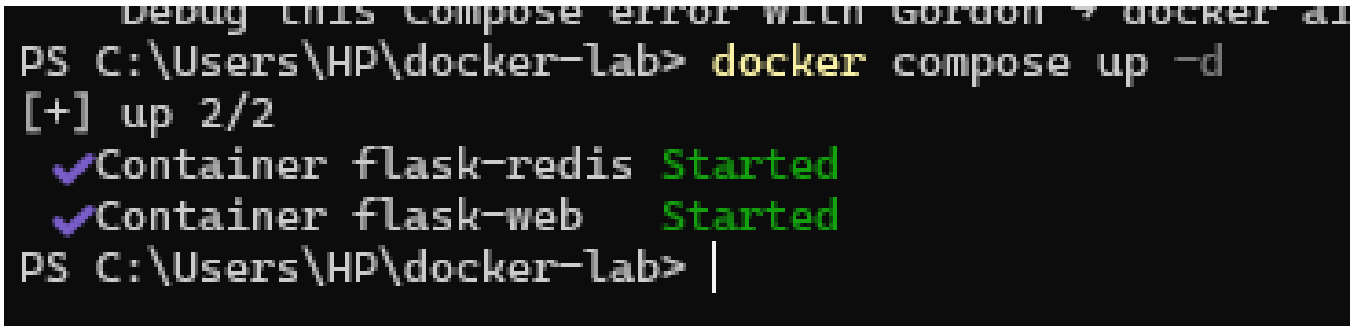
6. Docker Compose Configuration

```
PS C:\Users\HP\docker-lab> docker compose config
name: docker-lab
services:
  redis:
    container_name: flask-redis
    image: redis:7-alpine
    networks:
      default: null
  web:
    build:
      context: C:\Users\HP\docker-lab
      dockerfile: Dockerfile
    container_name: flask-web
    depends_on:
      redis:
        condition: service_started
        required: true
    networks:
      default: null
    ports:
      - mode: ingress
        target: 5000
        published: "5000"
        protocol: tcp
networks:
  default:
    name: docker-lab_default
```

7. Docker Compose Multi-Container Setup



8. Docker Compose Containers Running



9. Flask Application Verification

Hello! This application is running inside a Docker container.